

AI/ML Internship – Day 53 Notes & Tasks

Module 7: Stemming & Lemmatization – Reducing Words to Their Root Form

Part 1: Recap of Day 52

Yesterday we learned:

-  Text Preprocessing
-  Tokenization
-  Stop Words
-  Text Cleaning

Today we will learn:

-  Stemming
-  Lemmatization

These are very important NLP preprocessing techniques.

Part 2: Why Do We Need Stemming & Lemmatization?

In text data, the same word can appear in different forms.

Example:

play

playing

played

plays

All represent the same action.

If computers treat them as different words:

-  More complexity
-  Larger vocabulary
-  Poor model performance

So we convert them into a common root word.

✦ Part 3: What is Stemming?

Definition

Stemming is the process of removing prefixes or suffixes from a word to obtain its root form.

Example

Original Word Stemmed Word

Playing Play

Played Play

Plays Play

Running Run

Runner Run

Visual Representation

Playing

↓

Play

Played

↓

Play

Plays

↓

Play

✦ Part 4: How Stemming Works

Stemmers remove common endings:

ing

ed

s

es

ly

Example:

learning

↓

learn

working

↓

work

Part 5: Problems with Stemming

Sometimes stemming produces incorrect words.

Example:

studies

↓

studi

university

↓

univers

These are not proper English words.

Part 6: Stemming Using NLTK

Install NLTK:

```
pip install nltk
```

Code:

```
from nltk.stem import PorterStemmer
```

```
stemmer = PorterStemmer()
```

```
print(stemmer.stem("playing"))
```

```
print(stemmer.stem("played"))
```

```
print(stemmer.stem("plays"))
```

Output:

```
play
```

```
play
```

```
play
```

Part 7: What is Lemmatization?

Definition

Lemmatization converts words into their dictionary base form (Lemma).

Unlike stemming:

- ✓ Produces meaningful words
 - ✓ Uses language rules
 - ✓ More accurate
-

Example

Original Word Lemma

Playing	Play
---------	------

Played	Play
--------	------

Original Word Lemma

Better	Good
Running	Run
Mice	Mouse

🔴 Part 8: Stemming vs Lemmatization

Stemming	Lemmatization
Fast	Slower
Simple Rules	Uses Dictionary
Less Accurate	More Accurate
May Produce Invalid Words	Produces Valid Words

Example

Word:

studies

Stemming:

studi

Lemmatization:

study

🔴 Part 9: Lemmatization Using NLTK

Install required package:

```
pip install nltk
```

Download resources:

```
import nltk
```

```
nltk.download('wordnet')
```

Code:

```
from nltk.stem import WordNetLemmatizer
```

```
lemmatizer = WordNetLemmatizer()
```

```
print(
    lemmatizer.lemmatize("playing")
)
```

Output:

```
playing
```

To get proper result:

```
print(
    lemmatizer.lemmatize(
        "playing",
        pos="v"
    )
)
```

Output:

```
play
```

Part 10: More Lemmatization Examples

```
from nltk.stem import WordNetLemmatizer
```

```
lemmatizer = WordNetLemmatizer()
```

```
print(lemmatizer.lemmatize("cars"))
print(lemmatizer.lemmatize("mice"))
print(lemmatizer.lemmatize("children"))
```

Output:

car
mouse
child

Part 11: Real-World Example

Original Sentence:

The boys are playing footballs.

After Tokenization:

['The', 'boys', 'are', 'playing', 'footballs']

After Lemmatization:

['The', 'boy', 'are', 'play', 'football']

Now the text becomes more meaningful for machine learning.

Part 12: Why is Lemmatization Better?

Suppose we have:

run

running

runs

ran

Without Lemmatization:

4 different words

With Lemmatization:

run

One common root word.

This reduces vocabulary size.

📌 Part 13: NLP Workflow with Lemmatization

Raw Text
↓
Cleaning
↓
Tokenization
↓
Stop Word Removal
↓
Lemmatization
↓
Feature Extraction
↓
Model Training

📌 Part 14: Real-World Applications

Sentiment Analysis

Loved

Loving

Love

↓

Love

Chatbots

Better understanding of user messages.

Search Engines

Search:

running shoes

can match:

run shoes

Review Analysis

Reduces duplicate word variations.

Part 15: Complete Example

```
from nltk.stem import WordNetLemmatizer
```

```
lemmatizer = WordNetLemmatizer()
```

```
words = [  
    "cars",  
    "running",  
    "children",  
    "mice"  
]
```

```
for word in words:  
    print(  
        lemmatizer.lemmatize(word)  
    )
```

Output:

```
car  
running  
child  
mouse
```

Part 16: Theory Questions

What is Stemming?

Removing suffixes/prefixes to get root words.

What is Lemmatization?

Converting words into meaningful dictionary root forms.

Which is more accurate?

Lemmatization.

Why use Stemming and Lemmatization?

To reduce vocabulary size and improve NLP performance.

Give one example.

Playing → Play

Part 17: Interview Questions

Difference between Stemming and Lemmatization?

Stemming uses simple rules while Lemmatization uses dictionary-based language processing.

Which is preferred in modern NLP?

Lemmatization.

Why is Lemmatization important?

It improves text quality and model accuracy.

What is a Lemma?

Dictionary root form of a word.

Give an example where stemming fails.

Studies → Studi

Theory

1. What is Stemming?
 2. What is Lemmatization?
 3. Difference between Stemming and Lemmatization.
 4. Why is Lemmatization more accurate?
 5. Give three examples of root word conversion.
-

Practical Task 1

Install NLTK:

```
pip install nltk
```

Practical Task 2

Stem the following words:

Playing

Working

Learning

Running

Practical Task 3

Lemmatize:

Cars

Children

Mice

Dogs

Practical Task 4

Compare outputs of:

- Stemming
- Lemmatization

for the same words.

Practical Task 5

Create a table containing:

Word Stemmed Lemmatized

for 10 words.